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Air treatment system for recreational vehicles

Abstract

This present invention relates to a single body air treatment system for recreational vehicles and is a combination of components working in a scheme so to provide heating, cooling, dehumidification, ventilation, air-purification, heavy-duty drying, into a single body. The system comprising of a cooling air plant having a refrigerating cycle (**105,107,108,109**) and air heat exchangers acting with electricity and a heating plant having an incorporated fuel/Natural-Gas burner (**118,119**) acting as air furnace. The system operates heating and cooling simultaneously with in the same air-flow and provides air purification through an antibacterial electrical device. The system also integrates extra pipes to produce extra hot-water in winter while the system operates in heating. The system is integrated with a dedicated resonator chamber to minimize the external noise produce by extraction blowers.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates generally to the field of air treatment systems. More specifically,

the present invention relates to an air treatment system for recreational vehicles (RV) or trailers. The innovative design of air treatment system for recreational vehicle enables the system to be placed unconventionally in the lower part of the recreational vehicle or trailer, thereby not occupying any space outside the RV overall dimensions and is not visible outside. The air treatment system provides heating, cooling, dehumidification, ventilation, air purification, heavy-duty drying through a single integrated structure. Accordingly, this disclosure makes specific reference thereto the present invention. Nonetheless, it is to be appreciated that aspects of the present invention are also equally applicable to other like applications, devices and methods of manufacture.

Description of the Related Art

(2) Recreational vehicles often abbreviated as RV is a motor vehicle comprising of living quarters designed for accommodation. Families and retirees often use recreational vehicles to travel places nearby home by road, travel with more of the comforts of home, including pets, or a combination of both. Various types of RVs such as motorhomes, campervans, caravans (also known as travel trailers and camper trailers), fifth-wheel trailers, popup campers, truck campers or the like, are available in the market. The size of the recreational vehicles ranges from the size of being the size of a van all the way up to the size of a large bus.

(3) By way of background, recreational vehicle and trailers industry grew fast, improving quality, living space, comfort, aesthetic look or the like, however, very little innovation has been done in air-treatment devices for these recreational vehicles, which conceptually remain the same since decades. Usually, a plurality of air filters and air conditioner units are used for cleaning and cooling the air inside the recreational vehicle and to provide air ventilation for the users staying inside the recreational vehicle. The air conditioner units are usually heavy and are conventionally installed on a roof surface of the recreational vehicle, in order to save space inside the vehicle. However, weight of such multitude of air conditioners changes the center of gravity and other mechanical dynamics of the vehicle, and creates unbalancing and rocking instability.

(4) The existing air-conditioner technology requires an additional roof reinforcement to support the weight and shocks. The additional roof reinforcement increases the overall weight of the recreational vehicle. The additional weights increase the pulling force required to move the RV or trailer, thereby increase the fuel consumed for the operation of the vehicle.

(5) Additionally the heavy weight, air conditioners are quite large in size and creates resistance particularly to lateral winds, thereby exposing the vehicle to overturning. Moreover, the height of the existing air conditioner over the roof increases the possibility of accidents, damages or injuries caused by contact with trees, wire cables, wrapping with banners, collision with bridges and parking gates, or the like. This causes damage and loss to the user of the recreational vehicles, and requires the user to frequently maintain the vehicle for ready use.

(6) As part of the existing air conditioning technology, a large number of additional parts have to be installed indoor the vehicle to provide air distribution and proper sealing from rain and snow. Also, the current air conditioner technology requires people with extensive experience in the installation, trained maintenance and service which have to be provided working above the roof with safety risks. Further, the placement of the air conditioner units above the vehicle over the roof makes it difficult to service the system, as a person needs to climb the roof for servicing and maintenance of the air treatment system.

(7) Furthermore, different systems or devices are utilized for serving different purposes. For example, a heat furnace with a separate ducting is required for heating air and an air-cooling unit is separately installed for cooling the air. Traditionally, the heating is provided by a furnace, a gas burner powered by natural gas, such as LPG, propane, butane, located in a different place of the vehicle, more often inside the indoor furniture. Use of plurality of air treatment systems makes the complete vehicle heavy and causes problem in the operation of the vehicle. Also, due to use of multiple systems and their installation, space available inside the RV/Trailer is reduced, which increases installation costs and failure possibilities.

(8) Additionally, according to several environmental protection protocols and treaty, modern age technologies have raised strict limitations on the use of refrigerants such as Hydrochlorofluorocarbons (HCFCs), Hydrofluorocarbon (HFC), or the like. However, the technology used for air conditioner system in recreational vehicles is old and of the times when such limitations were not existing, and therefore, such air conditioner systems utilize large quantities of refrigerants, which leads to degradation of environment conditions. Also, the old technology-based conditioner systems have poor efficiency.
(9) The prior HVAC unit described in patent document US 2009/209193 is characterized by a number of drawbacks. In fact, this HVAC unit is designed in order to be integrated inside the lateral walls that internally delimit the living quarters of the recreational vehicle. Therefore, the air intake (the one from which the outside air is sucked inside the HVAC unit) is visible for people standing outside the recreational vehicle, with the consequence that the aesthetic appearance of the recreational vehicle (on which said HVAC unit is installed) is greatly damaged. Moreover, this HVAC unit can only work in a cooling or in a heating mode and therefore cannot provide dehumidification. If dehumidification is required, a separate dehumidification device must be provided.
(10) An apparatus for conditioning air inside a building is known from EP 1018624, which patent document teaches to provide a cooling air plant (condenser, evaporator, compressor) within a first space of a box body and to provide a heating air plant within a second space of the same box body separated from the first space. This apparatus is capable only of providing heating or cooling of the air and cannot in any way dehumidify air, thus requiring an additional dehumidification device if it is desired the option of dehumidifying.
(11) Therefore, there exists a long felt need in the art for an air treatment system for recreational vehicles which performs multiple functions such as cooling, heating, dehumidification, ventilation, etc. There is also a long felt need in the art for an air treatment system for recreational vehicles that can be easily installed on the vehicle without any assistance from an expert or trained person. There is also a long felt need in the art for an air treatment system for recreational vehicles which is light weight and small in size, therefore occupies less space of the recreational vehicle. There is also a long felt need in the art for an air treatment system for recreational vehicles that maintains the center of gravity of the vehicle and does not create unbalancing and rocking instability. Moreover, there is a long felt need in the art for an air treatment system for recreational vehicles that is installed in a position such that the system does not collides with trees, wires, etc., thereby preventing any damages caused to the vehicle and the user due to such collision. Also, there is a long felt need in the art for an air treatment system for recreational vehicles to be installed in a position such that the system is easily reachable and servicing can be performed conveniently. Additionally, there is a long felt need in the art for an air treatment system for recreational vehicles which comprises more than one system such as heating unit, cooling unit, or the like in a single integrated structure. Further, there is a long felt need in the art for an air treatment system for recreational vehicles that complies with the modern environmental protection protocols and treaty, and accordingly use the refrigerants in limited quantity. Finally, there is a long felt need in the art for an air treatment system for recreational vehicles that is efficient, light weight, cost effective and provides user convenience.
(12) The recreational vehicle according to the invention comprises a novel air treatment system which is a single body system comprising a cooling air module having a refrigerating cycle and air heat exchangers working on electricity, a heating module having an incorporated fuel/Natural-Gas burner acting as an air furnace, and a ventilation module to vent out the gases and heat.
(13) The subject matter disclosed and claimed herein, in one embodiment thereof, discloses a single metal sheet body air treatment system comprising a first chamber for providing air treatment, heating, cooling, dehumidification and air-purification, a second chamber to receive the return air, before treatment, a third chamber to remove extra heat absorbed by the system during the air-

cooling and a fourth chamber acting as a noise containment, with perforated shields and resonator to low the blower's sound.

(14) In this manner, the novel air treatment system of the present invention for recreational vehicles accomplishes all of the forgoing objectives, and provides a relatively environment friendly, easily installable, light weight, convenient and cost-effective solution to provide features such as air cooling, heating, ventilation, dehumidification, air purification and heavy duty drying by using a single integrated unit. The air treatment system of the present invention is also user friendly, inasmuch as the system is easily installable and serviceable, and do not require an expert or a skilled person to install the system onto the recreational vehicles.

SUMMARY OF THE INVENTION

(15) The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to present some general concepts in a simplified form as a prelude to the more detailed description that is presented later. The subject matter disclosed and claimed herein, in one embodiment thereof, discloses a novel air treatment system for recreational vehicles (RV) that is placed in the lower part of the RV and thus does not occupy space outside the RV overall dimensions and is also not visible from outside thus giving an aesthetic appeal to the RV. The novel air treatment system of the present invention provides heating, cooling, dehumidification, ventilation, air-purification and heavy-duty drying, all through a single body novel air treatment system. Further, a furnace required to produce heat is integrated into the air treatment system thus using components of the air treatment system.

(16) The novel air treatment system is a single body system comprising a cooling air module having a refrigerating cycle and air heat exchangers working on electricity, a heating module having an incorporated fuel/Natural-Gas burner acting as an air furnace, and a ventilation module to vent out the gases and heat.

(17) The subject matter disclosed and claimed herein, in one embodiment thereof, discloses a single metal sheet body air treatment system comprising a first chamber for providing air treatment, heating, cooling, dehumidification and air-purification, a second chamber to receive the return air, before treatment, a third chamber to remove extra heat absorbed by the system during the air-cooling and a fourth chamber acting as a noise containment, with perforated shields and resonator to low the blower's sound.

(18) The subject matter disclosed and claimed herein, in one embodiment thereof, discloses a recreational vehicle comprising an air treatment system placed in the lower section of the same recreational vehicle, wherein the air treatment system can provide air-cooling and air heating simultaneously. The combination of the fuel burner with air conditioning in the present invention controls indoor temperature, cold or hot, which is not possible with separated dehumidifiers currently available. During the air-cooling performed by the air treatment system, the incoming airflow condensate as much water as possible accordingly to the humidity/temperature conditions of the entering air making the airflow dry and cold, thereafter, the airflow is blown through a heat exchanger thus airflow absorbs heat and expands, thus becoming dried and comfortable. The apparatus eliminates dampness, humidity, molds (by means of an optional antibacterial electrical device), and can be used to dry cloths and furniture and provides a healthy environment even in rainy and wet days, all around the year.

(19) The present invention offers a new system and approach for air-conditioning in recreational vehicle design which does not occupy space outside the RV/Trailer overall dimensions and is not visible outside. The system provides heating, cooling, dehumidification, ventilation, air-purification, heavy-duty drying all in one module and body. The novel system can be easily installed and maintained from the ground, and does not require a user to climb the vehicle for repairing or replacement.

(20) The present invention improves vehicle stability reducing rocking and lower the center of gravity. Further, the system eliminates wind resistance reducing the risk of overturning. Since the system is not placed outside the vehicle or on the vehicle, the system eliminates the risk of collision with trees or wire cables banners, bridges or shrink parking gates and other similar objects.
(21) The system of the present invention is reducing cost and also allows for weight reduction as no additional roof-frames to support external air-conditioners. The components of the present invention can work using alternative energy sources such as solar panels with backup batteries and generators as well.
(22) To the accomplishment of the foregoing and related ends, certain illustrative aspects of the disclosed innovation are described herein in connection with the following description and the annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and are intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:
(2) FIG. **1** illustrates a schematic view of the metal body illustrating the distribution of main components of the air treatment system of the recreational vehicle of the present invention in accordance with the disclosed architecture;
(3) FIG. **2** illustrates a diagrammatic representation of the electrical distribution between the components of the air treatment system of the recreational vehicle of the present invention in accordance with the disclosed architecture;
(4) FIG. **3** illustrates a perspective view showing scheme of airflow distribution, airflow entry and exit in the air treatment system of the recreational vehicle of the present invention in accordance with the disclosed architecture;
(5) FIG. **4** illustrates an alternative scheme of airflow distribution, airflow entry and exit in the air treatment system of the recreational vehicle of the present invention in accordance with the disclosed architecture;
(6) FIG. **5** shows a schematic view of the single metal sheet body of the air treatment system of the recreational vehicle of the present invention illustrating the internal separation of the apparatus made of three main chamber and an optional fourth chamber;
(7) FIG. **6** illustrates a top perspective view of the single metal sheet body of the air treatment system of the recreational vehicle of the present invention in accordance with the disclosed architecture;
(8) FIG. **7** illustrates a bottom perspective view of the single metal sheet body of the air treatment system of the recreational vehicle of the of the present invention in accordance with the disclosed architecture;
(9) FIGS. **8**A and **8**B illustrates a schematic example illustrating an embodiment of the air treatment system capable also of producing extra hot water, such as for used in large RV's;
(10) FIG. **9** illustrates a schematic view of the placement of single metal sheet body of the air treatment system in the space available under the living quarters of the recreational vehicle;
(11) FIG. **10** illustrates a schematic section lateral view of a recreational vehicle according to the present invention, showing in particular the placement of the single metal sheet body of the air treatment system under the living quarters and the treatment air distribution system.
DETAILED DESCRIPTION OF THE SEVERAL EMBODIMENTS

(12) The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They are not intended as an exhaustive description of the invention and do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

(13) As noted above, there exists a long felt need in the art for an air treatment system for recreational vehicles which performs multiple functions such as cooling, heating, dehumidification, ventilation, etc. There is also a long felt need in the art for an air treatment system for recreational vehicles that can be easily installed on the vehicle without any assistance from an expert or trained person. There is also a long felt need in the art for an air treatment system for recreational vehicles which is light weight and small in size, therefore occupies less space of the recreational vehicle. There is also a long felt need in the art for an air treatment system for recreational vehicles that maintains the center of gravity of the vehicle and does not create unbalancing and rocking instability. Moreover, there is a long felt need in the art for an air treatment system for recreational vehicles that is installed in a position such that the system does not collides with trees, wires, etc., thereby preventing any damages caused to the vehicle and the user due to such collision. Also, there is a long felt need in the art for an air treatment system for recreational vehicles to be installed in a position such that the system is easily reachable and servicing can be performed conveniently. Additionally, there is a long felt need in the art for an air treatment system for recreational vehicles which comprises more than one system such as heating unit, cooling unit, or the like in a single integrated structure. Further, there is a long felt need in the art for an air treatment system for recreational vehicles that complies with the modern environmental protection protocols and treaty, and accordingly use the refrigerants in limited quantity. Finally, there is a long felt need in the art for an air treatment system for recreational vehicles that is efficient, light weight, cost effective and provides user convenience.

(14) Object of the present invention is a recreational vehicle. With the term “recreational vehicle” it is to be intended any motor vehicle comprising living quarters designed for accommodation of people or any trailer deemed to be attached to a car or van and comprising living quarters. Therefore, the terms “RV”, “trailers”, “motorhomes”, “campervans”, “caravans”, “fifthwheel trailers”, “popup campers” and “truck camper” are interchangeable and all falls within the definition of “recreational vehicle”.

(15) The recreational vehicle object of the present invention comprises living quarters **150** designed for accommodation of people and an air treatment system **100**, which comprises a single metal sheet body **101**.

(16) Referring initially to the drawings, FIG. **1** illustrates a schematic view of the single metal sheet body **101** illustrating the distribution of main components of the air treatment system **100** of the recreational vehicle of the present invention in accordance with the disclosed architecture. The components used in the present invention are all disposed into the single metal sheet body **101**, which can preferably be specifically treated to resist corrosion and shocks, acting as the containment. Electrical power supply to the components comprises an electrical power line socket **102**, a selector **103** and a power board **104**. The electrical power line socket **102** feeds the selector **103** which connects and transforms the power supply commonly available voltage (for example 12 Volts, 24 Volts, 115 Volts or 220 Volts) into the voltage required by the same air treatment system **100** to work (i.e. required by the components of the air treatment system **100** which will be

introduced in the description below). A remote switchboard (not shown) preferably provides the normal voltage to the powerline **102**.

(17) The single metal sheet body **101** is provided with internal separation in three main chambers (according to some embodiments, the chambers can be more than three, for example there can be an additional chamber). The single metal sheet body **101** comprises a first chamber **510** provided for conditioning a treatment air deemed to the living quarters **150**, a second chamber **520** provided for receiving a return air from the living quarters **150** and a third chamber **530** provided for treating a ventilation air to remove a heat absorbed by the same air treatment system **100**.

(18) In order to heat the treatment air, the air treatment system **100** comprises a heating air plant, which comprises a fuel or natural-gas burner assembly **118** for producing hot gases by means of a combustion. The heating air plant also comprises a sealed heat-exchanger **119**, which receives the hot gases from the burner assembly **118** and is placed within the first chamber **510** to safely exchange the heat of the hot gases through its surfaces (i.e. the surfaces of the sealed heat-exchanger **119**) with the treatment air for the living quarters **150**.

(19) In order to cool the treatment air, the air treatment system **100** comprises a refrigerator compressor **105** for the compression of a refrigerant fluid. The air treatment system **100** also comprises a condenser heat exchanger **107** that receives the compressed refrigerant fluid from the refrigerator compressor **105** through a first piping **106**. Therefore, in particular, during use, the refrigerator compressor **105** is provided for the compression of the refrigerant material into the condenser heat exchanger **107** through the first piping **106**. Also, the air treatment system **100** comprises a first blower **111** arranged for sucking the ventilation air from a first air intake **114**, for blowing it onto the condenser heat exchanger **107** and for expelling it outside of the single metal sheet body **101** of the air treatment system **100** through a first outlet **136** of the single metal sheet body **101**. Both the condenser heat exchanger **107** and the first blower **111** are placed into the third chamber **530**. Moreover, the air treatment system **100** comprises an evaporator heat exchanger **109** that receives the refrigerant fluid from the condenser heat exchanger **107** through an orifice tube **108** and absorbs the heat from the return air. Said evaporator heat exchanger **109** is placed in the first chamber **510**. Furthermore, the air treatment system **100** is provided with a second blower **113** placed within the second chamber **520** and arranged for sucking the return air from the living quarters **150** through a second air intake **112** of the single metal sheet body **101**, for blowing it on the evaporator heat exchanger **109** and on the sealed heat-exchanger **119** to produce the treatment air and for pushing said treatment air from the first chamber **510** to the living quarters **150** through at least one flange **115**. In particular, the second chamber **520** and the first chamber **510** communicate with each other (for example by means of an aperture provided on the internal separation of the single metal sheet body **101** between the first chamber **510** and the second chamber **520**), so that the second blower **113** in the second chamber **520** can blows the return air on the evaporator heat exchanger **109** and the sealed heat-exchanger **119**. Therefore, during use, from the condenser heat exchanger **107**, through an orifice tube **108**, the refrigerant fluid expands into the evaporator heat exchanger **109** crossed by the return air (which is transformed into the treatment air for the living quarters **150**) introduced into the air treatment system **100** through the second air intake **112** which is sucked in by the second blower **113**. During the expansion, the refrigerant reduces its temperature, thus, cooling the evaporator heat exchanger **109**. Also, the refrigerant fluid absorbs the heat from the return air and then the heated refrigerant returns to the refrigerator compressor **105** through a second piping **110**.

(20) The condenser heat exchanger **107** during the cooling operation is kept ventilated by the ventilation air generated by the first blower **111**. The ventilation air from the first blower **111** is required to cool the refrigerator compressor **105** and remove the heat from the condenser heat exchanger **107**. The exhausted ventilated air is expelled to the outside by means of the first blower **111**. During the crossing of the return air through the evaporator heat exchanger **109**, the return air gets refrigerated and dried (becoming thus the treatment air) into the first chamber **510**. From the

first chamber **510**, the refrigerated treatment air leaves the air treatment system **100** through flanges **115**, in the attached figures at the left extreme of the first chamber **510**, and becomes available for the air distribution.

(21) During an air-cooling process, the power board **104** supplies and controls the refrigerator compressor **105**, the first blower **111**, and the second blower **113**.

(22) Therefore, the power board **104** is preferably configured for executing a cooling mode by supplying and controlling the refrigerator compressor **105**, the first blower **111** and the second blower **113**. In this manner, during the cooling mode (i.e. during the air-cooling process), the first blower **111** sucks the ventilation air from the outside through the first air intake **114** and blows it against the condenser heat exchanger **107** and again to the outside of the third chamber **530**, the second blower **113** sucks a return air from the living quarters **150** and blows it against the evaporator heat exchanger **109** to transform said return air into the treatment air for the subsequent distribution to the living quarters **150**, while the refrigerator compressor **105** works so that the refrigerant material absorbs heat from the return air (transforming it into the treatment air) and discharges said heat to the ventilation air. Thus, after the air-cooling process, the return air is transformed into a cool treatment air for the living quarters **150**.

(23) The air treatment system **100** of the present invention for heating the air can operate in a heating mode, during which in particular an air-heating process is performed. The apparatus **100** integrates a fuel or natural-gas burner assembly **118**, arranged for burning fuel or natural gas. The combustible fuel is provided to the burner assembly **118** in particular through a third fitting **116** and its pipe which feeds a valve **117** which supplies the combustible fuel or natural gas. The burner assembly **118** includes sensors and electrodes and fuel orifice. The combustible fuel (or natural gas) then flows through the orifice before and get ignited. The combustible fuel (or natural gas) burns into the sealed heat-exchanger **119**, which is advantageously positioned in line after the burner assembly **118**. The power board **104** controls the entire burning process that takes place in the air treatment system **100**. The sealed heat-exchanger **119** acts as burning chamber and heat exchanger at the same time. The function of the sealed heat-exchanger **119** is to safely exchange the heat generated by the combustion through its surfaces into the first chamber **510** (in the figures, present on the left extreme) where the heat gets released. From the first chamber **510**, the heated treatment air leaves the air treatment system **100** through the flanges **115** (at the left extreme of the first chamber **510**) and becomes available for the air distribution.

(24) The sealed heat-exchanger **119** internally defines a combustion chamber isolated by the first chamber **510** (i.e. isolated by the return air/treatment air in order to avoid that combustion gases could pollute the return air/treatment air that has to be introduced into the living quarters **150**). Thus, the sealed heat-exchanger **119** requires a dedicated air-flow to operate the internal combustion. To provide the proper quantity of air/oxygen required for the combustion, the heating air plant comprises a third blower **121** arranged for feeding the combustion chamber of the sealed heat-exchanger **119** with an air-flow.

(25) During use, the third blower **121** forces air, in particular through the burner assembly **118**, into the sealed heat-exchanger **119** to make combustion possible and also to blow out exhaust gasses through an extension pipe **122**, which advantageously communicates with the combustion chamber of the sealed heat-exchanger and departs from said sealed heat-exchanger **119**. The extension pipe **122** and its external flange **123** are made of stainless steel to extend the lifespan of the air treatment system **100**. During an air-heating process, the power board **104** supplies and controls the second blower **113** and the heating air plant.

(26) Therefore, the power board **104** is preferably configured for executing a heating mode by supplying and controlling the second blower **113** and the heating air plant. In this manner, during the heating mode (i.e. during the air-heating process), the second blower **113** sucks a return air from the living quarters **150** and blows it against the sealed heat-exchanger **119** to transform said return air into the treatment air for the subsequent distribution to the living quarters **150**, while the

burner assembly **118** executes a combustion and the sealed heat-exchanger **119** transfers through its surfaces the heat produced by the combustion to the return air for transforming the latter into an hot treatment air.

(27) Preferably, during said air-heating process, the power board **104** supplies and controls at least the second blower **113**, the valve **117**, the burner assembly **118** and the third blower **121**. Therefore, the power board **104** is preferably configured for executing a heating mode by supplying and controlling the second blower **113** and the valve **117**, the burning assembly **118** and the third blower **121** of the heating air plant. In this manner, during the heating mode, the second blower **113** sucks a return air from the living quarters **150** and blows it against the sealed heat-exchanger **119** to transform said return air into the treatment air for the subsequent distribution to the living quarters **150**, while the valve **117** opens to supply with combustible fuel or natural gas the burner assembly **118**, the burner assembly **118** ignites said combustible fuel or natural gas and the third blower **121** blows an air-flow through the same the burner assembly **118** and inside the combustion chamber of the sealed heat-exchanger **119** to render possible the combustion inside said sealed heat-exchanger **119**.

(28) The temperatures of the exhausted gases, the water and the acids produced during the combustion by the air treatment system **100** could damage the same system and, therefore, the extension pipe **122** and its external flange **123** are made of stainless steel to extend the lifespan of the apparatus **100**.

(29) FIG. **2** illustrates a diagrammatic representation of the electrical distribution between the components of the air treatment system **100** of the present invention in accordance with the disclosed architecture. As previously mentioned, the air treatment system **100** comprises an electrical power supply, which comprises an electrical power line socket **102**, a selector **103** for connecting and transforming the commonly available voltage into the voltage required by the air treatment system **100** to work and a power board **104** arranged for supplying and controlling at least the refrigerator compressor **105**, the first blower **111**, the second blower **113** and the heating air plant. Thus, during use, the electrical power line socket **102** feeds the selector **103** which connects and transforms the power supply commonly available 12 Volts, 24 Volts, 115 Volts or 220 Volts into the voltage required by the air treatment system **100** to work. The electrical power line socket **102**, the selector **103**, the power board **104** and the remote switchboard **127** provide the necessary electrical power for the components to work for cooling the air, heating the air, dehumidifying the air (as will be described below) and indoor ventilation. A dehumidification mode (which will be described below), the cooling mode and heating mode can be selected by a user through a remote switchboard **127**. Thus, the air treatment system **100** advantageously comprises a remote switchboard **127** by means of which a user can control the functioning of the same air treatment system **100**, for example by selecting one from the group comprising at least the cooling mode, the heating mode and the dehumidification mode. The power-board **104** supplies and controls the refrigerator compressor **105**, the blower **111**, the second blower **113**. The power board **104** also controls the entire burning process.

(30) Advantageously, the air treatment system **100** comprises an antibacterial electrical device **126** to eliminate airborne spores produced by plants, fungi and molds and purify treatment air for the living quarters **150** from most of airborne bacteria. More in detail, the antibacterial electrical device **126** is placed inside the first chamber **510**. In particular, as can be seen from FIG. **1** for example, said antibacterial electrical device **126** is placed between the second blower **113**, on the one hand, and the evaporator heat exchanger **109** and the sealed heat-exchanger **119**, on the other hand (for example in front of the aperture provided on the internal separation of the single metal sheet body **101** between the first chamber **510** and the second chamber **520**). In this manner, the antibacterial electrical device **126** operates on the return air before the latter is transformed into treatment air for the living quarters **150** by the evaporator heat exchanger **109** and/or the sealed heat-exchanger **119**.

(31) As shown in FIG. **2**, the electrical power is supplied from power board **104** to various

components such as thermostat **120**, third blower **121**, antibacterial electrical device **126**, burner assembly **118** through electrical connections. The electrical distribution between the components of the air treatment system **100** of the present invention is supplied in the voltage required for the components to work. It should be appreciated that any component used in the single metal sheet body **101** of the present invention which require electrical power would be provided the power through the power-board **104** which takes the electrical power from remote switchboard **127**. The remote switchboard **127** is to be wired from the power-board into the RV/Trailer where users can operate temperatures, ventilation speeds and functions. The remote switchboard **127** can be operated manually or remotely by mean of wi-fi applications through a wireless communication technology such as Wi-Fi, Bluetooth, NFC, Infrared and the like. The remote switchboard **127** may be operated using a smartphone application remotely. Thus, the remote switchboard **127** preferably comprises a wireless communication module for being operated remotely by mean of wi-fi applications.
(32) The thermostat **120** works as a safety device to interrupt combustible fuel or natural gas supply and avoid overheating in case of faults.
(33) The single metal sheet body **101** of the air treatment system **100** is placed under the living quarters **150**. In this manner, the single metal sheet body **101** (and therefore the whole air treatment system **100**) is at least almost entirely hidden from the sight of the people staying in the living quarters **150** and does not reduce or occupy space deemed for the accommodation of people.
(34) Preferably, the recreational vehicle according to the invention comprises a treatment air distribution system **140**, which connects the at least one flange **115** of the first chamber **510** with the living quarters **150** for distributing the treatment air from the first chamber **510** to the living quarters **150**.
(35) In particular, the treatment air distribution system **140** is made of distribution ducts departing from the flanges **115** of the first chamber **510** and ending in emission mouths **141** in the living quarters **150**.
(36) As the single metal sheet body **101** is placed under the living quarters **150** and the recreational vehicle comprises a treatment air distribution system **140**, the air treatment system **100** is hidden from the view of people, no space inside the living quarters **150** or inside the furniture of the same living quarters **150** is occupied by the system **100** and no huge mouths for blowing the treatment air inside the living quarters **150** or for expelling the ventilation air outside of the recreational vehicle (which could be seen from the inside or from the outside of the recreational vehicle) are needed. Indeed, the distribution ducts, which depart from the first chamber **510** (in particular from the flanges **115**), can extend below the floor **137** of the living quarters **150** and/or inside the vertical walls delimiting the living quarters **150** of the recreational vehicles, so that a plurality of small (and therefore hardly visible) emission mouths **141** for the treatment air can be provided on the floor **137** of the living quarters **150** and on the vertical walls that delimit the living quarters **150**.
(37) Moreover, as preferably the treatment air distribution system **140** comprises a plurality of hidden distribution ducts that end in corresponding emission mouths **141** on the floor **137** or on the vertical walls of the living quarters **150**, the flowing of the treatment air inside the living quarters **150** is particularly uniform because the treatment air, in this manner, comes from a plurality of different points distributed in the living quarters **150**.
(38) As can be seen from the annexed figures, the single metal sheet body **101** is preferably provided with a top wall **138**, with an opposite bottom wall **135** and with a lateral wall **139** connecting the top wall **138** and the bottom wall **135**.
(39) In particular, the single metal sheet body **101** is placed under the living quarters **150** with its top wall **138** substantially parallel to and facing the floor **137** of the living quarters **150**.
(40) More in detail, the first air intake **114** is provided on the bottom wall **135** of the third chamber of the single metal sheet body **101**. This configuration of the first air intake **114** renders the single metal sheet body **101** (and therefore the air treatment system **100**) particularly suitable for being

placed under the living quarters **150**. Indeed, in this manner, the ventilation air can be sucked by the first blower **111** from the space between the lower portion of the recreational vehicle and the road surface. The first air intake **114** is therefore not visible for people standing outside the recreational vehicle and avoids that huge and aesthetically not appealing intake apertures are placed on the sides of the recreational vehicle.
(41) Furthermore, the second air intake **112** is advantageously provided on the top wall **138** of the second chamber **520** of the single metal sheet body **101**.
(42) The position of this second air intake **112** renders particularly easy the sucking of the return air by means of the second blower **113** for example by simply providing an opening (and eventually a corresponding duct) on the floor **137** of the living quarters **150** above the second air intake **112**. This configuration of the second air intake **112** therefore renders the single metal sheet body **101** (and therefore the air treatment system **100**) particularly suitable for being placed under the living quarters **150**.
(43) Moreover, the at least one flange **115** is provided on the top wall **138** and/or on the lateral wall **139** of the first chamber **510** of the single metal sheet body **101**. In particular, the provision of the flanges **115** on the top wall **138** and/or on the lateral wall **139** renders the installation of the air treatment system **100** under the living quarters **150** particularly easy. For example, this configuration of the flanges **115** permit to reduce the overall length of the distribution ducts of the treatment air distribution system **140**.
(44) FIG. **3** illustrates a perspective view showing scheme of treatment, return and ventilation air distribution, air entry and exit in the air treatment system **100** of the present invention in accordance with the disclosed architecture. As shown in the FIG. **3**, the flanges **115** are used to vent out the refrigerated treatment air for the treatment air distribution system **140**. Similarly, the heated treatment air leaves the air treatment system **100** through the flanges **115** and becomes available for the treatment air distribution system **140**. It should be appreciated that flanges **115** installed on the main body **101** can be multiple and in various direction to simplify the air treatment system **100** installation and efficiency. The return air enters from the second air intake **112** passes through the second blower **113**, crosses the evaporator heat exchanger **109** and the sealed heat-exchanger **119** before leaving the air treatment system **100** through the flanges **115** and becoming available for the treatment air distribution system **140**. Preferably, the air treatment system **100** comprises a dripping pipe **124**. More in detail, as can be seen in FIGS. **3** and **4**, said dripping pipe **124** is in particular placed in communication with the interior of the first chamber **510**, starts from the bottom wall **135** of the single metal sheet body **101** and extends outside of the same first chamber **510**. Thus, the condensate water extracted during the cooling of the air is drip outside the apparatus **100** through the dripping pipe **124**.
(45) Preferably, the single metal sheet body **101** comprises a fourth chamber **540** operating as a resonator to reduce outside noise of the first blower **111** and, furthermore, the air treatment system **100** comprises punched shields **134** present inside said fourth chamber **540** to help in dying sound waves to reduce noise. This fourth chamber **540** is optional and can be seen for example in the embodiment of FIGS. **1**, **2**, **3**, **5**, **6** and **7**, while it is not provided in the embodiment of FIG. **4**.
(46) As shown, punched shields **134** (used to remove noise) are present in the right most chamber (i.e. the fourth chamber **540**) of the air treatment system **100**, which chamber operates as a resonator, in which the sound waves reflect against each other and loose energy by mean of punched shields **134**.
(47) The ventilation air enters from the first air intake **114**, flows around the refrigerator compressor **105**, then through the condenser heat exchanger **107** and get expelled to the outside by the first blower **111** through the first outlet **136**, as shown in FIG. **3**. As can be seen for example in FIGS. **3** and **5**, the third chamber **530** and the fourth chamber **540** communicates with each other (by means of the first outlet **136** provided on the internal separation of the single metal sheet body **101** between the third chamber **530** and the optional fourth chamber **540**), so that the first blower

111 expels the ventilation air to the outside through the first outlet **136** and through the fourth chamber **540**, which opens towards the outside (differently, if the fourth chamber **540** is not present, the third chamber **530** directly opens towards the outside by means of the first outlet **136**, in order to discharge the ventilation air blown by the first blower **111**).

(48) FIG. **4** illustrates an alternative scheme of airflow distribution, airflow entry and exit in the air treatment system **100** of the present invention in accordance with the disclosed architecture. In the present embodiment, the apparatus **100** does not have the right most chamber (i.e. the fourth chamber **540**), which works as a resonator, and in this embodiment the ventilation air is directly expelled to the outside by means of the first blower **111**. The length and size of the system **100** of the present embodiment (without fourth chamber **540**) is shorter thus, and it is beneficial for smaller RVs with less space and also in which less noise is created.

(49) FIG. **5** shows a schematic view of the single metal sheet body **101** of the air treatment system **100** of the recreational vehicle according to the present invention illustrating the internal separation of the apparatus made of three main chambers **510**, **520**, **530** and an optional fourth chamber **540**. As shown, first chamber **510** is responsible for the air treatment, heating, cooling, dehumidification and air-purification. The first chamber **510** also has multiple flanges **115** to vent out both cold and hot treatment air.

(50) A second chamber **520** receives the return air, before treatment. A third chamber **530** contains the components in charge to remove the extra heat absorbed by the air treatment system **100** during the air-cooling. An optional fourth chamber **540** acts as a noise containment, with perforated shields (i.e. punched shields **134**) and resonators to low the first blower's **111** extraction noise for user convenience. The single metal sheet body **101** of the air treatment system **100** as well as the components used in the present invention are designed to be shock-proof, compatible with water splashes and the impact of little stones, as per automotive good practice. Further, all internal components are suspended inside to allow a fast and quick inspection of the air treatment system **100** from below by just removing a lower panel **135** (which is the bottom wall **135**) of the body **101**. One additional advantage that the present invention offers is that the lower panel **135** is also replaceable in case of damages or oxidations to increase the life of the air treatment system **100** and internal components. The entire lower panel **135** can also be lowered and removed for inspection, cleaning, maintenance or replacement of the components.

(51) FIG. **6** illustrates a top perspective view of the air treatment apparatus body of the present invention in accordance with the disclosed architecture. From the top view, the second air intake **112**, electrical power line socket **102**, flanges **115** and third fitting **116** are visible. The combustible is provided to the burner assembly **118** through the third fitting **116**. The return air enters into the single metal sheet body **101** from the second air intake **112**. It should be appreciated that the single metal sheet body **101** is closed from the top such that the efficiency and protection of the components are maintained.

(52) FIG. **7** illustrates a bottom perspective view of the air treatment apparatus body of the present invention in accordance with the disclosed architecture. A dripping pipe **124** is visible which is used to drip the condensate water extracted during the cooling of the air. The entire lower panel **135** is visible of the body **101** in the bottom perspective view of the body **101**. From the bottom, first air intake **114** is shown from which ventilation air is entered, which ventilation air is expelled outside through the first blower **111**.

(53) According to a particular embodiment shown in FIGS. **8**A and **8**B, the air treatment system **100** comprises an optional additional heat-exchanger **128**, which is placed in the first chamber **510** after the sealed heat-exchanger **119** along the trajectory followed by the treatment air blown by the second blower **113** and is arranged for producing extra hot water. Thus, during a heating mode, this additional heat-exchanger **128** absorbs heat from the sealed heat exchanger **119**. More in detail, the recreational vehicle comprises a secondary closed circuit for a coolant, wherein the additional heat-exchanger **128** is part of said secondary closed circuit and is arranged for receiving a coolant from

a first fitting **129** (preferably placed on the top wall **138**, as can be seen from FIG. **6**) and for discharging said coolant from a second fitting **130** (preferably placed on the top wall **138**, as can be seen from FIG. **6**). Thus, the air treatment system **100** comprises a first fitting **129**, through which the coolant enters the additional heat-exchanger **128**, for example when pushed by the pump **132** (described below), and a second fitting **130**, through which the coolant leaves the additional heat-exchanger **128**, for being directed for example to a water tank (in this case the coolant must be water) or to a plate-heat-exchanger **131** (described below—in this case the coolant can be also a different fluid instead of water). Therefore, during a heating mode (or air-heating process), a cold coolant enters the additional heat-exchanger **128** through the first fitting **129**, flows within said additional heat-exchanger **128** absorbing heat from the sealed heat-exchanger **119** and exits from the additional heat-exchanger **128** by means of the second fitting **130**. The coolant could be for example water, which increases its temperature by flowing through the additional heat exchanger **128** during a heating mode and exits through the second fitting **130**. The water as coolant can be directed from the second fitting **130** to the inside of a water tank (which is therefore part of the secondary closed circuit).

(54) Differently, the recreational vehicle comprises a plate-heat-exchanger **131**, placed inside said living quarters **150** and the secondary closed circuit is placed to connect the plate-heat-exchanger **131**, and the additional heat-exchanger **128** and is intercepted by a pump **132** arranged for circulating the coolant inside the secondary closed circuit between the plate-heat-exchanger **131** and the additional heat-exchanger **128**. In this variant, the coolant is preferably an antifreeze and not water and the plate-heat-exchanger **131** is for example placed to heat a water tank.

(55) Thus, the additional heat-exchanger **128** is placed in the extreme left chamber (i.e. the first chamber **510**) in the same airflow generated by the second blower **113** after the sealed heat-exchanger **119**. The additional heat-exchanger **128** is placed to produce extra hot water in winter particularly in large RV's where hot-water demand can be increased by the number of people or when water tanks may chill down in very cold environment.

(56) FIGS. **8**A and **8**B illustrates a schematic example illustrating said particular embodiment for producing extra hot water by the air treatment system **100** such as for used in large RV's. This present embodiment gives the advantage of producing extra hot water in winter particularly in large RV's where hot-water demand can be increased by the number of people or when water tanks present in the RV may chill down in very cold environment. The present embodiment builds a secondary closed circuit using antifreeze as transfer heat fluid instead of water which results also better to prevent oxidations and freezing. As shown in FIG. **8**B, an additional heat-exchanger **128** is placed in the first chamber **510** in the same air-flow generated by the blower **113** after the heat-exchanger **119**. In cold environments the usage of antifreeze as transfer heat is furthermore recommended to protect the heat-exchanger **128** when the air treatment system **100** is not working. The coolant enters from the first fitting **129** by mean of the pump **132** and leaves from the second fitting **130** to reach a plate-heat-exchanger **131** located inside the RV/Trailer. In FIG. **8**A, an expansion/refill tank **133** is also preferably part of the secondary closed circuit. The plate-heat-exchanger **131** can be connected accordingly to satisfy various needs and desires of the user. The present embodiment may be used to provide hot water to store into an insulated tank or keeping existing water tanks protected from freezing.

(57) FIG. **9** illustrates a schematic view of the placement of single metal sheet body **101** of the air treatment system **100** in the space available under the living quarters **150** of the RV/Trailer. Advantageously, the living quarters **150** are below delimited by a floor **137** and the single metal sheet body is placed within a storage compartment **910** provided under said floor **137**. Preferably, the recreational vehicle comprises structural frames **900** under the living quarters **150** and the single metal sheet body **101** of the air treatment system **100** can be placed in a storage compartment **910** defined between said structural frames **900**. This embodiment with the single metal sheet body **101** placed in a storage compartment **910** defined between structural frames **900** allows to use for

the same single metal sheet body **101** a space that difficulty could be used for other scopes and that is therefore generally underexploited. In particular, the structural frames **900** comprise longitudinal structural frames **900**, which are substantially parallel to the advancement direction of the recreational vehicle, and transversal structural frames **900**, which are substantially orthogonal to the longitudinal structural frames **900** to define storage compartments **910** that are, in plant view, substantially rectangular (in FIG. **9**, there are depicted on the left the structural frames **900** of a recreational vehicle in form of trailer and on the right the structural frames **900** of a recreational vehicle in form of a motor vehicle).

(58) The power board **104** is arrangeable to make the heating air plant and the cooling air plant to work simultaneously, so that the evaporator heat exchanger **109** dries the return air and the sealed burner **119** heats the return air received from the evaporator heat exchanger **109** to produce said treatment air dried and hot.

(59) Therefore, the power board **104** is preferably configured for executing a dehumidification mode by supplying and controlling the refrigerator compressor **105**, the first blower **111**, the second blower **113** and the heating air plant. In this manner, during the dehumidification mode: the first blower **111** sucks the ventilation air from the outside through the first air intake **114** and blows it against the condenser heat exchanger **107** and again to the outside of the third chamber **530**, the second blower **113** sucks a return air from the living quarters **150** and blows it firstly against the evaporator heat exchanger **109** in order to cool down the return air and to condensate the water vapor carried within said return air and secondly against the sealed heat exchanger **119** in order to reheat said cool and dry treatment air and transforming it into a hot and dried treatment air, the sealed heat-exchanger **119** transfers heat generated by the combustion to the return air cooled down and dried by the evaporator heat exchanger **109**, and the refrigerator compressor **105** works so that the refrigerant fluid absorbs heat from the return air by flowing through the evaporator heat exchanger **109** and discharges said heat to the ventilation air by flowing through the condenser heat exchanger **107**.

(60) Thus, after the dehumidification mode, the return air is transformed into a dry and hot treatment air for the living quarters **150**.

(61) Thus, the air treatment system **100** of the present invention has the ability to operate heating and cooling simultaneously with in the same return air, even controlling temperature cold or hot, thus becoming an effective drier all year around to eliminate damp, humidity, molds, dry cloths and furniture's and provide a healthy environment even in rainy and wet days.

(62) As previously stated, the air treatment system **100** advantageously comprises an antibacterial electrical device **126**. More in detail, this antibacterial electrical device **126** is placed inside the first chamber **510**, so that it can operate on the return air/treatment air. In particular, the antibacterial device **126** is placed between the second blower **113**, on the one hand, and the sealed heat-exchanger **119** and the evaporator heat exchanger **109**, on the other hand. Thus, additionally, air purification is obtained by said antibacterial electrical device **126** capable to eliminate airborne spores produced by plants, fungi and molds and therefore capable to purify the return air/treatment air from most of airborne bacteria. Moreover, the system may integrate extra pipes (i.e. the additional heat-exchanger **128**) to produce extra hot-water in winter while the air treatment system **100** operates in heating mode. Finally, the air treatment system **100** may be integrated with a dedicated resonator chamber (i.e. the fourth chamber **540**) to minimize the external noise produce by extraction blowers (i.e. the first blower **111**).

(63) The air treatment system **100** of the present invention is a combination of components working in a scheme to provide Heating, Cooling, Dehumidification, Ventilation, Air-Purification, heavy-Duty Drying, into a single body, which is thus easy to install, maintain, replace and maintain. Further, the air treatment system **100** is lightweight, consumes less space and is installed in the lower section of RV thus does not reduce the appearance of the RV/Trailer.

(64) The cooling air plant of the air treatment system **100** has a refrigerating cycle air heat

exchangers (i.e. the condenser heat exchanger **107** and the evaporator heat exchanger **109**) acting with electricity (as the refrigerator compressor **105** is supplied by the power board **104**). The heating air plant has an incorporated fuel/Natural-Gas burner **118** acting as air furnace.
(65) The air treatment system **100** of the recreational vehicle of the present invention reduces the quantity of the refrigerant fluid used for cooling, thus is environmentally friendly, which may affect the global warming and reconsidered a new way of miniaturization to spare space and raw materials. Alternative sources such as solar energy, wind energy, generators may be used. The single metal sheet body **101** of the present invention is designed to be placed unconventionally in the lower part of the recreational vehicle according to the invention. The various size of the existing vehicles or trailers impose a minimum and a maximum apparatus thickness to respect the distance from the ground and possibly fit between the structural frames **900**. The inner components are sized to satisfy air cooling needs in the range from one to three times the today's most popular standards, taking as reference one roof-top ac's, so eliminating a multitude of airconditioners as used nowadays.
(66) Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not structure or function.
(67) Notwithstanding the forgoing, the air treatment system **100** of the recreational vehicle of the present invention can be of any suitable size and configuration as is known in the art without affecting the overall concept of the invention, provided that it accomplishes the above stated objectives. One of ordinary skill in the art will appreciate that the size, configuration and material of the air treatment system **100** as shown in the FIGS, are for illustrative purposes only, and that many other sizes and shapes of the air treatment system **100** are well within the scope of the present disclosure. Although the dimensions of the air treatment system **100** are important design parameters for user convenience, the air treatment system **100** may be of any size that ensures optimal performance during use and/or that suits the user's needs and/or preferences.
(68) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. While the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.
(69) What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are possible. Accordingly, the claimed subject matter is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

Claims

1. A recreational vehicle comprising: living quarters designed for accommodation of people; an air treatment system comprising: single metal sheet body provided with internal separation in at least three main chambers; wherein a first chamber is provided for conditioning a treatment air deemed to the living quarters, a second chamber is provided for receiving a return air from the living

quarters before treatment and a third chamber is provided for treating a ventilation air; a heating air plant comprising: a fuel or natural-gas burner assembly for producing hot gases by means of a combustion process; a sealed heat-exchanger which receives the hot gases from said burner assembly and is placed within said first chamber to exchange the heat of the hot gas through its surfaces with the treatment air for the living quarters; a cooling air plant comprising: a refrigerator compressor for the compression of a refrigerant fluid; a condenser heat exchanger that receives the compressed refrigerant fluid from said refrigerator compressor through a first piping; a first blower arranged for sucking said ventilation air from a first intake of said single metal sheet body, for blowing it onto said condenser heat exchanger and for expelling it to the outside of the single metal sheet body of the air treatment system through a first outlet of said single metal sheet body; said condenser heat exchanger and said first blower being placed into said third chamber; an evaporator heat exchanger that receives the refrigerant fluid from said condenser heat exchanger through an orifice tube and absorbs the heat from said return air; said evaporator heat exchanger being placed within said first chamber; a second blower placed within the second chamber and arranged for sucking the return air from the living quarters through a second air intake of said single metal sheet body, for blowing it into said first chamber on said evaporator heat exchanger and on said sealed heat-exchanger to produce said treatment air and for pushing said treatment air from said first chamber to the living quarters through at least one flange; an electrical power supply comprising a power line socket, a selector for connecting and transforming an available voltage into a voltage required by the air treatment system to work and a power board arranged for supplying and controlling at least the refrigerator compressor, the first blower, the second blower and the heating air plant; said power board being arrangeable to make the heating air plant and the cooling air plant to work simultaneously, so that the evaporator heat exchanger dries the return air and the sealed heat-exchanger heats the return air received from the evaporator heat exchanger to produce said treatment air dried and hot; said single metal sheet body being placed under the living quarters; said single metal sheet body comprising a fourth chamber operating as a resonator to reduce outside noise of the first blower; said air treatment system comprising punched shields present inside the fourth chamber to help in dying sound waves to reduce noise.

2. The recreational vehicle of claim 1, wherein said living quarters are below delimited by a floor; the single metal sheet body being placed within a storage compartment provided under said floor.

3. The recreational vehicle of claim 2, further comprising structural frames under the living quarters; the single metal sheet body of said air treatment system being placed in said storage compartment defined between said structural frames.

4. The recreational vehicle of claim 1, wherein the single metal sheet body of said air treatment system is provided with a top wall, with an opposite bottom wall, and with a lateral wall connecting the top wall and the bottom wall; said first air intake being provided on the bottom wall of the third chamber of said single metal sheet body.

5. The recreational vehicle of claim 4, wherein said second air intake is provided on the top wall of the second chamber of said single metal sheet body.

6. The recreational vehicle of claim 4, wherein said at least one flange is provided on the top wall and/or on the lateral wall of the first chamber of said single metal sheet body; said recreational vehicle comprising a treatment air distribution system, which connects said at least one flange of said first chamber with the living quarters for distributing said treatment air from said first chamber to the living quarters.

7. The recreational vehicle of claim 4, wherein said air treatment system comprises a dripping pipe, which is placed in communication with the interior of said first chamber, starts from the bottom wall of said single metal sheet body and extends outside of said first chamber.

8. The recreational vehicle of claim 1, wherein said second blower is located upstream of said evaporator heat exchanger, which is placed between said second blower and said sealed heat-exchanger so that said second blower forces the return air to pass first through said evaporator heat

exchanger and then through said sealed heat-exchanger.
9. The recreational vehicle of claim 8, wherein said sealed heat-exchanger comprises a shaped tube conveying the hot gases and placed in front of said evaporator heat exchanger.
10. The recreational vehicle of claim 1, wherein said sealed heat-exchanger internally defines a combustion chamber isolated from the first chamber; said heating air plant comprising a third blower arranged for feeding the combustion chamber of said sealed heat-exchanger with an air-flow.
11. The recreational vehicle of claim 1, wherein the said air treatment system comprises a remote switchboard by means of which a user can control the functioning of the same air treatment system.
12. The recreational vehicle of claim 11, wherein said remote switchboard comprises a wireless communication module for being operated remotely by mean of wi-fi applications.
13. A recreational vehicle comprising: living quarters designed for accommodation of people; an air treatment system comprising: single metal sheet body provided with internal separation in at least three main chambers; wherein a first chamber is provided for conditioning a treatment air deemed to the living quarters, a second chamber is provided for receiving a return air from the living quarters before treatment and a third chamber is provided for treating a ventilation air; a heating air plant comprising: a fuel or natural-gas burner assembly for producing hot gases by means of a combustion process; a sealed heat-exchanger which receives the hot gases from said burner assembly and is placed within said first chamber to exchange the heat of the hot gas through its surfaces with the treatment air for the living quarters; a cooling air plant comprising: a refrigerator compressor for the compression of a refrigerant fluid; a condenser heat exchanger that receives the compressed refrigerant fluid from said refrigerator compressor through a first piping; a first blower arranged for sucking said ventilation air from a first intake of said single metal sheet body, for blowing it onto said condenser heat exchanger and for expelling it to the outside of the single metal sheet body of the air treatment system through a first outlet of said single metal sheet body; said condenser heat exchanger and said first blower being placed into said third chamber; an evaporator heat exchanger that receives the refrigerant fluid from said condenser heat exchanger through an orifice tube and absorbs the heat from said return air; said evaporator heat exchanger being placed within said first chamber; a second blower placed within the second chamber and arranged for sucking the return air from the living quarters through a second air intake of said single metal sheet body, for blowing it into said first chamber on said evaporator heat exchanger and on said sealed heat-exchanger to produce said treatment air and for pushing said treatment air from said first chamber to the living quarters through at least one flange; an electrical power supply comprising a power line socket, a selector for connecting and transforming an available voltage into a voltage required by the air treatment system to work and a power board arranged for supplying and controlling at least the refrigerator compressor, the first blower, the second blower and the heating air plant; said power board being arrangeable to make the heating air plant and the cooling air plant to work simultaneously, so that the evaporator heat exchanger dries the return air and the sealed heat-exchanger heats the return air received from the evaporator heat exchanger to produce said treatment air dried and hot; said single metal sheet body being placed under the living quarters; wherein said air treatment system comprises an additional heat-exchanger, which is placed in the first chamber after the sealed heat-exchanger, along a trajectory followed by said treatment air blown by said second blower, and is arranged for producing hot water.
14. The recreational vehicle of claim 13, further comprising a secondary closed circuit for a coolant; said additional heat-exchanger being part of said secondary closed circuit and arranged for receiving a coolant from a first fitting and for discharging said coolant from a second fitting.
15. The recreational vehicle of claim 14, further comprising a plate-heat-exchanger placed inside said living quarters; said secondary closed circuit being placed to connect said plate-heat-exchanger and said additional heat-exchanger and being intercepted by a pump arranged for circulating said coolant inside said secondary closed circuit between the plate-heat-exchanger and the additional

heat-exchanger.
16. The recreational vehicle of claim 15, wherein the coolant is an antifreeze.